Leading Agentic AI Transformation in the Organization

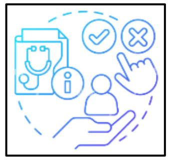
Agent

- A software system that can
  - ✓ Perceive its environment
  - ✓ Reason about it
  - ✓ Make decisions
  - ✓ Take actions to achieve specific goals. Unlike traditional programs that follow fixed rules,

- Use advanced techniques
  - ✓ Machine Learning
  - ✓ Natural Language Processing (NLP)
  - ✓ Decision-making algorithms.

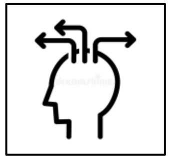
Agents: Key Characteristics

Goal-Driven – Oriented toward completing tasks or optimizing outcomes



Autonomy – Operate without constant human supervision

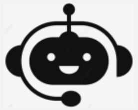
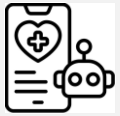
Reactivity – Sense and respond to changes in their environment.



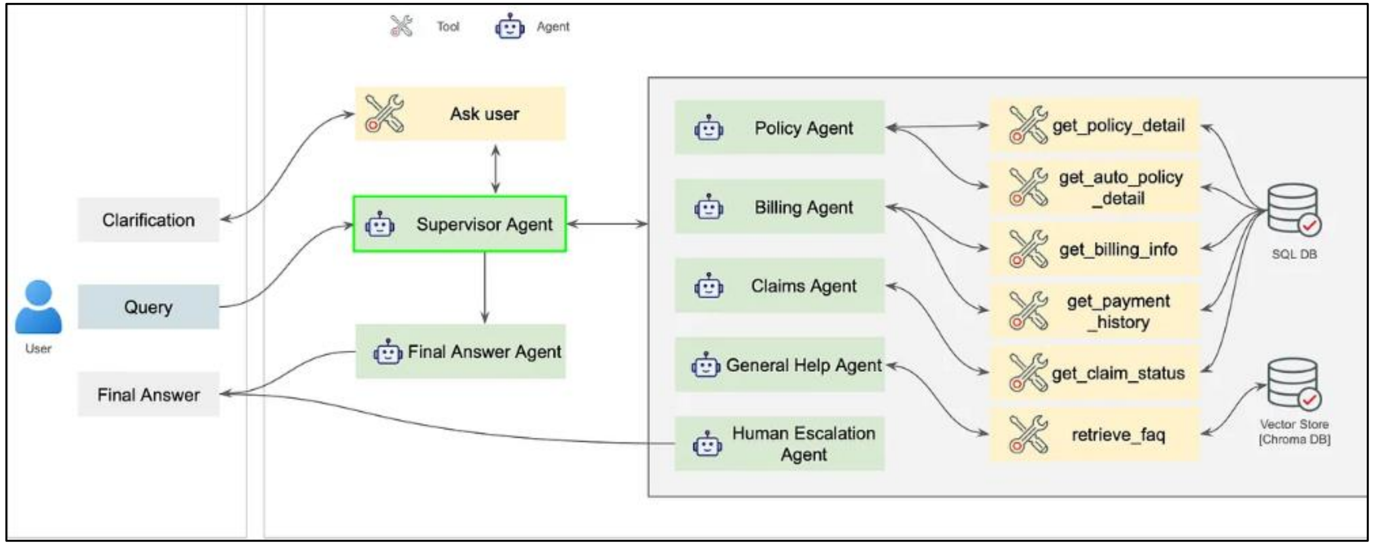
Proactivity – Take initiative to achieve predefined goals.

Learning – Improve performance over time by analyzing data and feedback.

Agents – Use Cases

	**Customer Support Chatbot (E-commerce)**	An AI chatbot that handles customer inquiries, processes orders, checks inventory, and resolves complaints in real-time
	**Automated Research Assistant (Finance & Investment)**	A financial research bot that fetches live Stock market data, analyzes trends, and provides investment insights.
	**Smart IT Helpdesk Agent (IT & DevOps)**	An AI assistant for troubleshooting system issues, fetching logs, and suggesting fixes
	**Personalized Legal Document Generator (Legal & Compliance)**	A legal AI assistant that drafts contracts, NDA agreements, or compliance documents based on real-time inputs
	**AI-powered Medical Assistant (Healthcare)**	A virtual assistant that helps doctors with patient diagnosis, retrieves medical records, and provides drug recommendations.

Multi Agents



- A system composed of multiple autonomous entities (**agents)**
- Interact with each other to perform tasks, solve problems, or achieve specific goals.
- Each agent operates independently, has specialized capabilities, and can communicate or collaborate with other agents to complete complex objectives.
- Coordinate to handle tasks requiring diverse expertise, iterative reasoning, or decision-making processes.
- Every agent has Its own role, goal, and backstory
- Access to different tools or knowledge

Multi Agents – Key Features

Decentralization	No single agent controls the entire system; decisions are made collectively
Scalability	New agents can be added to the system without redesigning the entire architecture
Flexibility	Agents can be reassigned or removed based on task requirements
Robustness	The system can continue functioning even if some agents fail
Specialization	Each agent focuses on one domain (support, sales, analytics

Agentic AI

- AI systems that can
  - ✓ Act independently
  - ✓ Make decisions
  - ✓ Complete tasks

with minimal human intervention

- These AI systems have a sense of "agency"
  - ✓ They can set goals
  - ✓ Plan, adapt, and execute actions based on their environment or data they process.

Agentic AI Foundational Capabilities

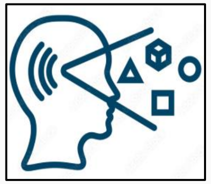
Capability	Description
Autonomy	Agent operates independently with minimum human intervention **Example:** Schedule a meeting Automatically checks calendars for free slots, send invite
Reasoning	Thinking and planning capability of the Agent **Example:** Convert a large document into multiple chunks
Action	Connect Decision with Outcome. **Example:** which Tool to invoke after a certain action

Traditional Automation vs Agentic AI

Factors	Traditional Automation	Agentic AI
Definition	Rule Based	Perceive, Reason, Plan and achieve its goal
Flexibility	No. Restricted by conditions	Adaptive: responds to change in inputs and outputs
Decision Making	Usually deterministic	Probabilistic
Handling Uncertainty	Not good. Overfitting occurs	Good. Can interpret ambiguity well
Reinforcement Learning	No. Static.	Yes. Learn from prompt adaptation and fine tuning
Best suited for	Repetitive tasks	Complex, dynamic situations

Components of Agentic AI

Goals



Perception

Reasoning

Plan

Act

Feedback

Memory

Agentic AI – Examples

	**Virtual Personal Assistants**	They can schedule meetings, set reminders, and even order groceries online based on your instructions and preferences.
	**Self-Driving Cars**	These cars make decisions on the road, such as when to slow down, change lanes, or stop at signals, all without human input.
	**Customer Support Chatbots**	Intelligent bots that can resolve customer issues, escalate problems when necessary, and learn from previous interactions to improve future responses.
	**AI-powered Trading Bots**	Bots that analyze stock market data in real time and automatically execute trades based on market conditions to maximize profit.

Design Patterns

Traditional Software Engineering → Design Pattern → Reusable Solution to commonly occurring problems

Agentic AI System → Design Pattern → How autonomous AI agents think, plan, collaborate, remember, decide, and act to solve complex tasks.

Why Agentic AI Design Patterns Matter

- AI systems become
  - ✓ Modular
  - ✓ Scalable
  - ✓ Reliable
  - ✓ Explainable
  - ✓ Autonomous

- With Design Patterns
  - ✓ Responsibilities are separated
  - ✓ Reasoning becomes traceable
  - ✓ Systems become maintainable
  - ✓ Multiple agents can collaborate effectively

Common Design Patterns in Agentic AI

- ReAct
- Tool-Using
- Hierarchical
- Planner Executor
- Retrieval-Augmented
- Multi-Agent Collaborator
- Event Driven
- Reflection
- Human-In-The-Loop
- Memory Augmented

Decision Frameworks

<u>Agentic vs Automation vs Application Development</u>

- Three distinct ways of executing work
  - ✓ 3.1 Deterministic systems
  - ✓ 3.2 RPA (Robotic Process Automation) represent
  - ✓ 3.3 Autonomous systems
- From fully reasoning agents to strict rule-based automation.
- Choosing the right approach depends on parameters like variability, ambiguity, autonomy needs, and system complexity.
- Real-world scenarios map cleanly to these frameworks, helping teams decide whether a task needs
  - ✓ **Thinking**
  - ✓ **Structured Logic** or
  - ✓ **Repeatable Automation**

Choosing the Right Approach

- Choosing the right approach depends on
  - ✓ Nature of the business problem
  - ✓ Level of decision-making required
  - ✓ Data complexity
  - ✓ Adaptability needs.

Deterministic Systems

- When processes are fixed, predictable, and rule-based.

RPA

- When repetitive human tasks across applications need automation without changing existing systems.

Autonomous/Agentic Systems

- When the application must reason, adapt, make decisions, and handle dynamic or unstructured scenarios.

- *The best enterprise solutions often combine all three approaches to achieve reliability, automation, and intelligent decision-making together*.

Building production ready Agentic Apps

A. Challenges in production deployment
B. Hallucination handling and reliability
C. Cost optimization strategies
D. Prompt engineering for agents
E. Agent evaluation and testing approaches
F. Performance tuning and scalability
G. Security and responsible AI considerations
H. Deployment architectures and enterprise patterns

End